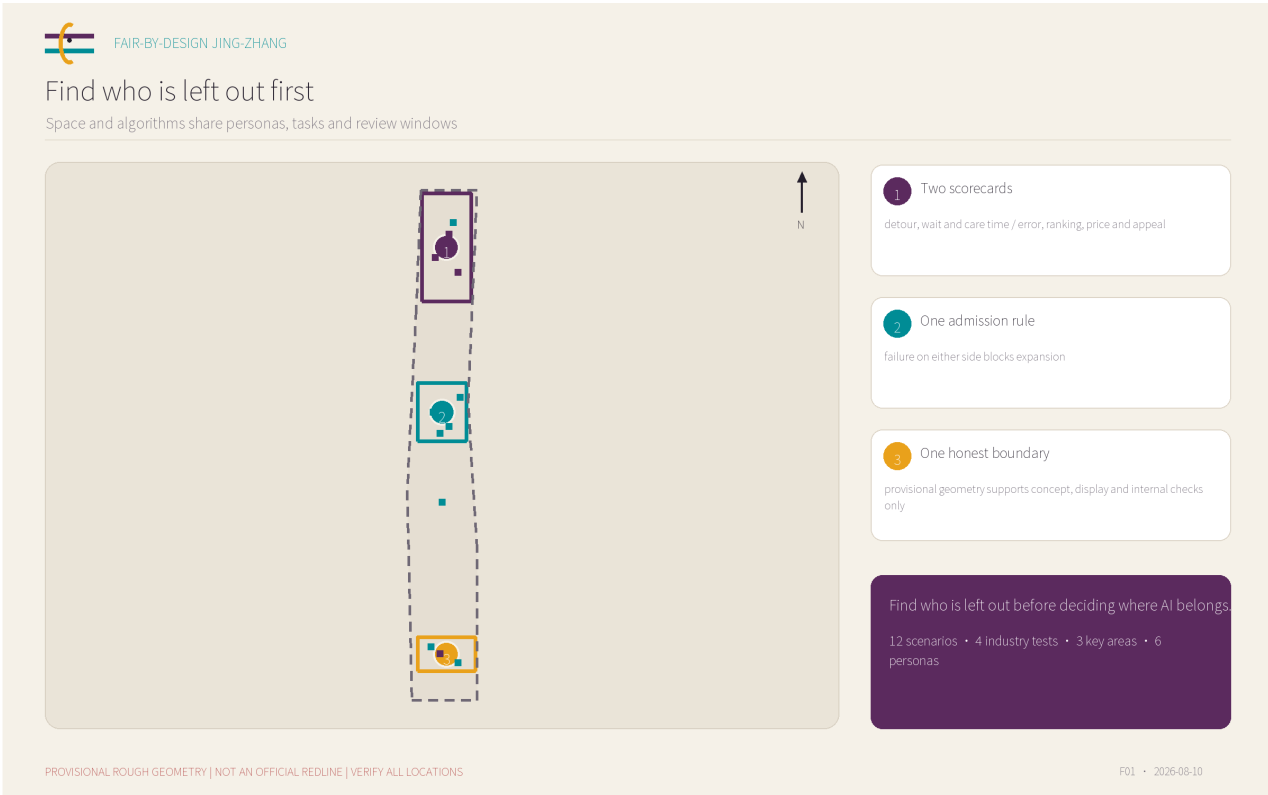
FAIR-BY-DESIGN JING-ZHANG: A Dual Calibration Protocol for Space and Algorithms

Find who is left out before deciding where AI belongs.

Fair-by-design is a continuous calibration procedure, not a zero-bias claim. Six personas, twelve scenarios and three key areas require space and algorithms to pass together. Every location uses provisional geometry.

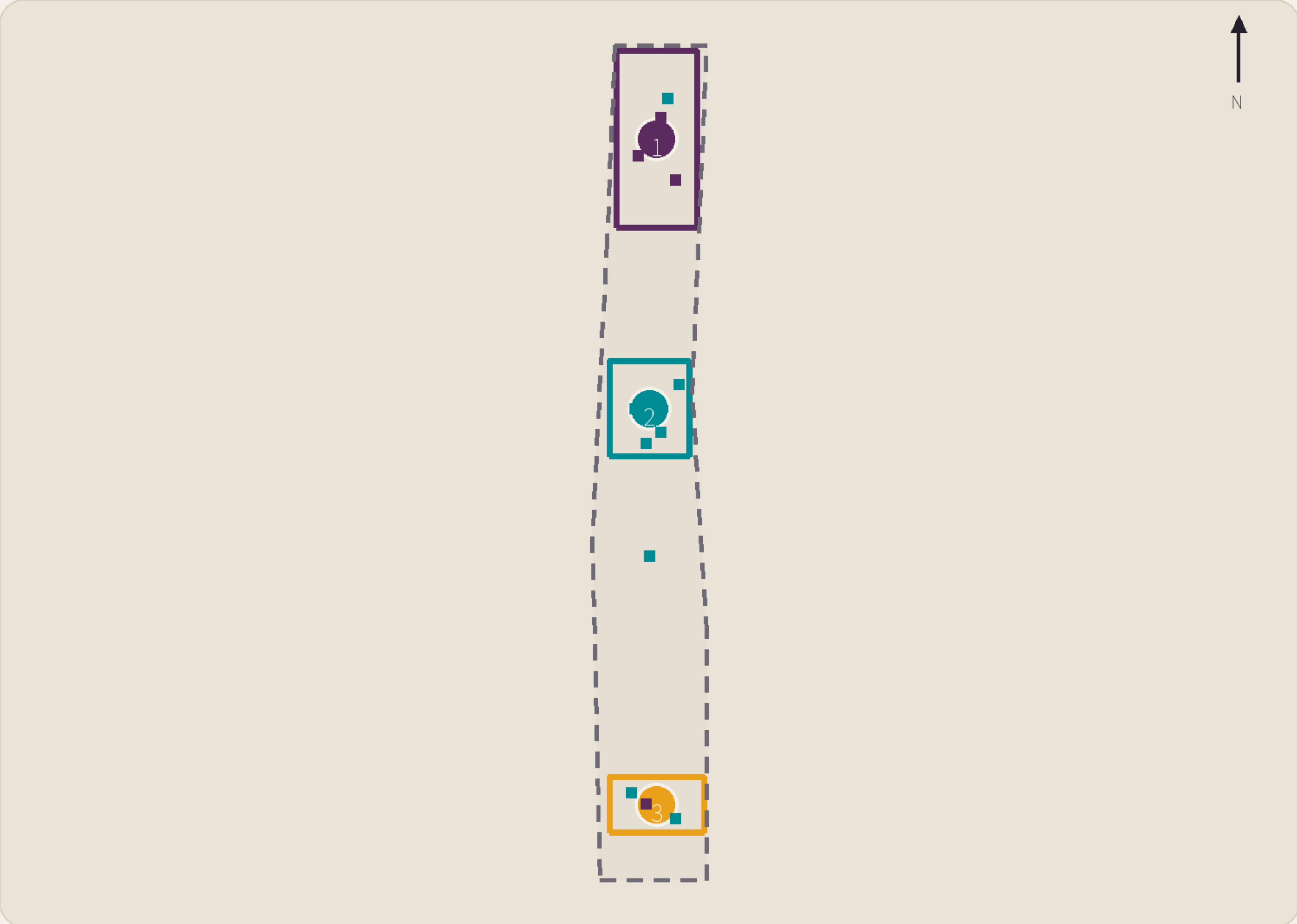


01 Concept and boundary

FAIR-BY-DESIGN JING-ZHANG

Find who is left out first

Space and algorithms share personas, tasks and review windows



1 Two scorecards

detour, wait and care time / error, ranking, price and appeal

2 One admission rule

failure on either side blocks expansion

3 One honest boundary

provisional geometry supports concept, display and internal checks only

Find who is left out before deciding where AI belongs.

12 scenarios • 4 industry tests • 3 key areas • 6 personas

PROVISIONAL ROUGH GEOMETRY | NOT AN OFFICIAL REDLINE | VERIFY ALL LOCATIONS

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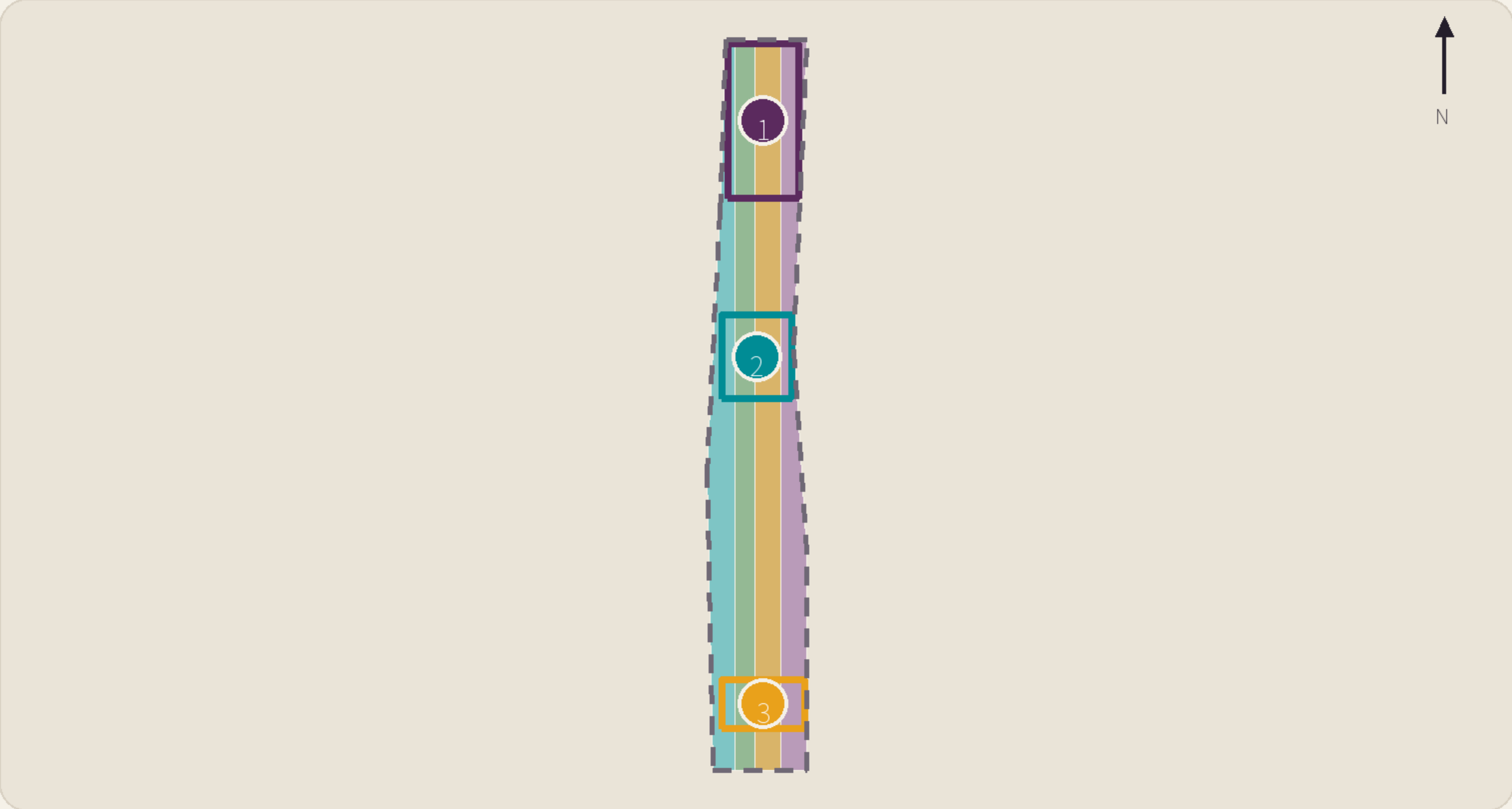
02 Scope and spatial structure

SIX DAILY SECTIONS

Six daily sections, not another smart axis

Codes are preserved; conceptual functions are not statutory zoning



- Research / fair testing
- Blue-green / culture
- Industry service / daily test
- Care, learning / community

COMMON CALIBRATION SCALE

Access	Care
Learning	Work
Service	Voice

Baseline → reversible trial → measure gaps → human review → public retrospective → expand / modify / withdraw

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03 Three key areas

THREE TESTING GRADIENTS

Three key areas: test - co-design - daily parity

Each key area has a verified, correctable and withdrawable honour node

Zhongzhiyuan: Bias Stress-Test Yard

Test before entering live streets. Closed courses, mirrored-data rooms, adjustable sections and a public results wall examine model error and spatial exclusion together.

SCENARIOS

- T1 Mirrored CV and venture-matching test
- T2 Speech assistant equal-error test
- T3 Robot yielding and care-time test
- S5 Inclusive lab access and PPE fit

HONOUR NODE Fairness Ruler

AI Origin Community: Care-Chain Prototype

Link work, learning, childcare, eldercare, health and rest as a walkable time chain, while testing whether booking and allocation impose extra time on carers.

SCENARIOS

- S6 Fair care-booking and waiting
- S7 Maternal and family health navigation boundary test
- S8 Equal-seat launch and contribution credit
- S9 Education and career recommendation calibration

HONOUR NODE Equal-Seat Forum

Dazhongsi: Night-Fairness Interface

Improve night travel through light, sightlines, active frontage and staffed help - never facial recognition as a substitute for design - while auditing routes, transfers, ranking, price and rostering.

SCENARIOS

- T4 Night-route parity test
- S10 Automated shuttle parity experience
- S11 Commercial recommendation and price consistency audit
- S12 Dual-bias appeal and public retest

HONOUR NODE Care Transfer Arcade

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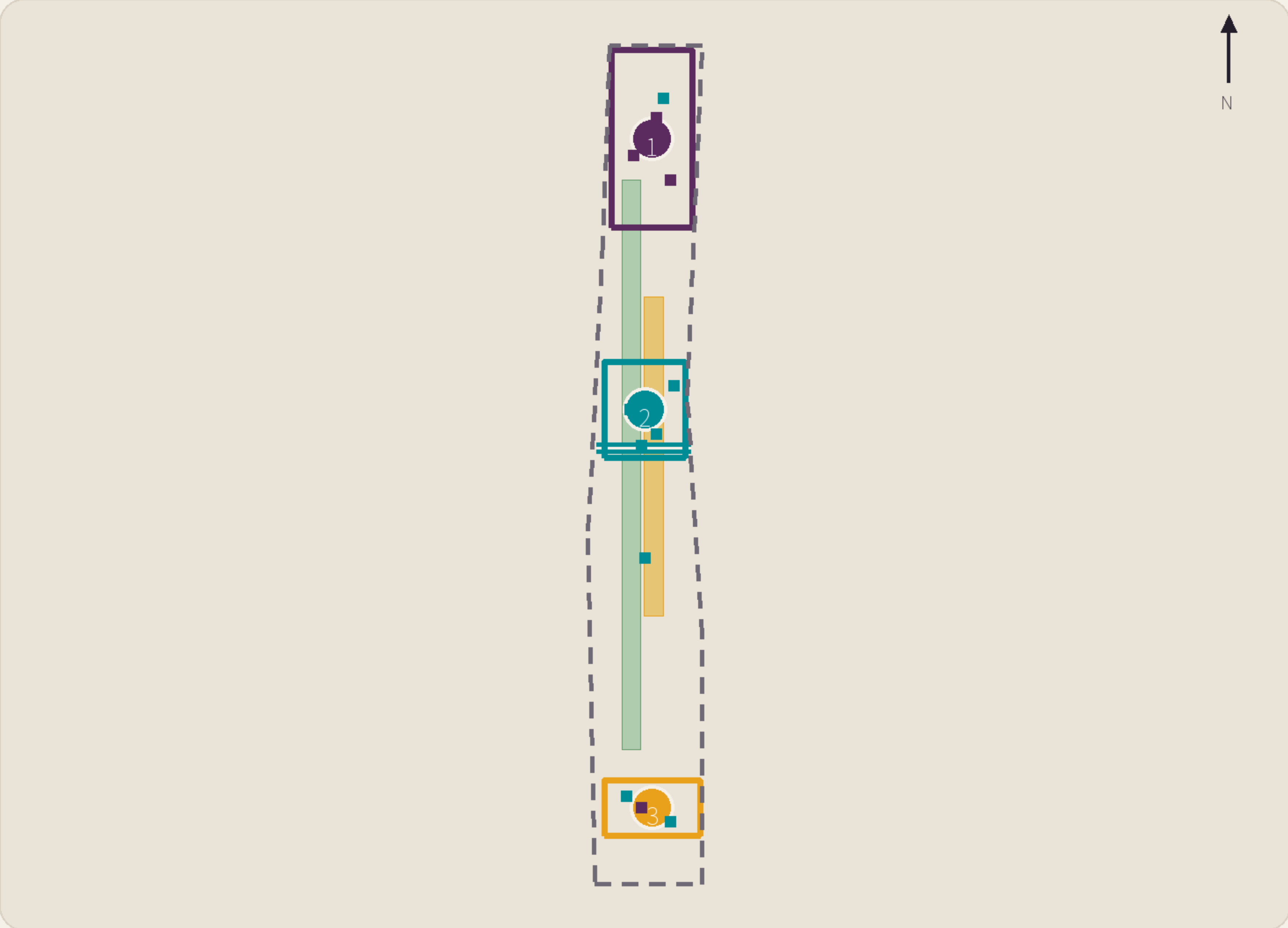
04 Mobility, night travel and blue-green space

WELL LIT, NOT IDENTIFIED

Equal access at night: repair space before testing AI

Robots, shuttles and navigation retain emergency stop, staffed degradation and offline service



1 Clear width

wheelchair, stroller and slow-walker passing

2 Night lighting

continuous, low-glare, no identity recognition

3 Last service

time, cost and false advice

4 Loading

paper/staff first, then assisted scheduling

5 Robot yielding

zero contact, disclosed near misses, physical stop

6 Three-channel wayfinding

digital, paper/tactile and staffed

PROVISIONAL ROUGH GEOMETRY | NOT AN OFFICIAL REDLINE | VERIFY ALL LOCATIONS

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05 Metrics, evidence and stop gates

EVIDENCE BEFORE PRECISION

Known, provisional and unknown - no false precision

D30 establishes baselines; insufficient samples never become a pass

RECALCULABLE / KNOWN

- provisional site area 11.413 km^2
- 12 scenario nodes
- 4 industry tests
- 3 honour nodes

PROVISIONAL

- green and public-space ratios
- concept building footprint
- three key-area polygons
- phasing and scenario locations

UNKNOWN

- FAR, height and density
- road, rail and utility controls
- local group differences
- operator, budget and permission

SPATIAL SCORECARD

S1	detour / wait / clear width
S2	lighting / seat / toilet
S3	care time / staffed help

ALGORITHMIC SCORECARD

A1	error / rejection / ranking
A2	price / allocation / explanation
A3	appeal outcome / rollback

FAIL EITHER SIDE → DO NOT EXPAND | SERIOUS SAFETY/PRIVACY EVENT → STOP

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Six test personas

ID	Persona	Task	Risk
P1	woman AI founder working late	a predictable chain between funding, hiring, labs and late transport	ranking bias, night detours, and portraying women only as protected users
P2	male engineer with primary childcare duties	stroller access, childcare booking, flexible events and child-welcoming work space	care being reassigned to women, or booking systems penalising atypical schedules
P3	pregnant or breastfeeding postdoctoral researcher	lab safety, fitting PPE, health navigation, rest and return-to-work continuity	leakage of sensitive health data and indirect exclusion by access or rostering systems
P4	older woman without a smartphone who walks slowly	legible wayfinding, frequent rest, toilets, staffed help and paper proof	combined exclusion by digital identity, default walking speed and speech errors
P5	night-shift cleaning, retail or delivery worker	visible rest points, night interchange, safe loading and appealable algorithmic scheduling	serving visitors while overlooking essential workers, and replacing spatial safety with surveillance
P6	visitor using a wheelchair, mobility aid or stroller	continuous clear width, gentle gradients, entrance choice, robot yielding and accessible event seating	accessible routes becoming longer second-class routes, or robots passing only idealised tests

Personas expose blind spots; they are not demographics. Sensitive identity must never be inferred from faces, voices, behaviour or routes.

Twelve dual-scorecard scenarios • 1-3

T1 • Mirrored CV and venture-matching test

Carrier/groups	controlled evaluation room and public results wall in the Zhongzhiyuan bias stress-test yard; women founders, carers and researchers across ages
Space	difference in access to equal-size booths, equal time slots and accessible pitching seats
Algorithm	group gaps in selection, ranking and explanation completeness for qualification-matched mirrored CVs
Data/privacy	synthetic CVs, volunteer ratings and a human benchmark; no live recruitment database; synthetic identities only; publish aggregate results, never individual scores
Owner/baseline/window	yard operator plus independent employment and algorithm reviewers; two pre-launch mirrored baselines, with sample size preregistered by a statistician; 30/60/90 d
Success	all mirrored-group gaps stay within preregistered tolerance and 100% receive readable explanations
Stop	any group breaches tolerance twice, rankings cannot be explained, or real personal data appears

T2 • Speech assistant equal-error test

Carrier/groups	noise-adjustable meeting booths with text input and human note-taking; speakers across pitch, accent, age and assistive-device use
Space	device reach and task-time gaps across stature and hearing needs
Algorithm	group gaps in word error, command completion and correction burden
Data/privacy	consented volunteer recordings, synthetic speech and human transcripts; no voiceprint identification; delete raw audio on schedule; text-only participation available
Owner/baseline/window	test-yard operator plus linguistics and accessibility reviewers; separate human and model baselines by noise level and interaction mode; 30/60/90 d
Success	completion gaps meet preregistered tolerance without degrading fallback completion
Stop	misrecognition causes a high-risk command, recordings cannot be deleted, or text fallback fails

T3 • Robot yielding and care-time test

Carrier/groups	a low-speed closed course with variable clear widths, separate from live streets; wheelchair, walker and stroller users across stature and speed
Space	minimum clearance, detour, waiting time and parity of accessible routes
Algorithm	group gaps in detection, yielding, near misses and human takeover
Data/privacy	closed-course event logs, human observation and anonymous volunteer feedback; no facial or gender inference; video retained briefly for redacted safety review only
Owner/baseline/window	robot provider, independent transport-safety supervisor and public observers; human cart and no-robot runs form same-route controls; 30/60/90 d
Success	zero collisions; near-miss and group-yielding gaps below preregistered thresholds
Stop	one collision, clustered near misses, failed emergency stop, or blocked clear width

Twelve dual-scorecard scenarios • 4-6

T4 • Night-route parity test

Carrier/groups	two comparable walking segments at the Dazhongsi night-fairness interface; night workers, women founders and slow or mobility-aid users
Space	gaps in time, cost, lighting continuity, active frontage, rest and last-service connection
Algorithm	group gaps in route ranking, cost estimation, interchange feasibility and false alerts
Data/privacy	public timetables, manual lux readings, volunteer walk logs and observation; no faces or inferred gender; one-time route IDs and aggregate publication
Owner/baseline/window	district operator, transport and lighting reviewers, and worker representatives; record human-recommended and existing digital routes at matching times; 30/60/90 d
Success	no group is systematically routed farther, darker, or past the last service
Stop	a route cannot be completed safely, last-service advice fails, or surveillance replaces spatial repair

S5 • Inclusive lab access and PPE fit

Carrier/groups	height-adjustable access prototype, PPE fitting zone and staffed visitor desk; pregnant or breastfeeding researchers, short-stature and wheelchair users, and visitors
Space	reach, clear width, PPE size coverage and queue-time gaps
Algorithm	group gaps in rejection, false alarms and human override time
Data/privacy	synthetic credentials, manual access checks and voluntary fit feedback; no pregnancy or health labels; credentials and feedback stored separately
Owner/baseline/window	lab safety lead and accessibility adviser; time existing staffed access in parallel with the prototype; 30/60/90 d
Success	all participants pass independently or receive staffed help within two minutes
Stop	lock-in, requested health status, or admission with clearly ill-fitting PPE

S6 • Fair care-booking and waiting

Carrier/groups	care advice desk and quiet waiting area in the AI Origin equal-seat commons; single-parent households, male carers, older people and people without smartphones
Space	gaps in walk distance, seats, toilets, stroller parking and waiting
Algorithm	group gaps in booking, waitlist ranking, reminder reach and no-show penalties
Data/privacy	test slots, synthetic household situations, and paper and phone task logs; no real child data; conduct a separate sensitive-data assessment before live service
Owner/baseline/window	community operator, care providers and public reviewers; compare app, phone and paper channels against identical test capacity; 30/60/90 d
Success	channel completion and wait gaps fall within preregistered tolerance
Stop	phone-less users cannot complete, care status is commercialised, or waitlist logic is opaque

Twelve dual-scorecard scenarios • 7-9

S7 • Maternal and family health navigation boundary test

Carrier/groups	private consultation room, public service-directory wall and staffed referral desk; pregnant and breastfeeding people, partner-carers and families across ages
Space	privacy, accessibility, referral walk time and companion seating
Algorithm	information completeness, mis-referral and escalation of high-risk questions
Data/privacy	public service directories and human-edited answers; no medical records; no diagnosis or retained health queries; clear sessions on exit
Owner/baseline/window	public-service navigator and clinical-content reviewer; staffed directory-search accuracy and completion time form the baseline; 30/60/90 d
Success	directory accuracy meets the human-reviewed target and all high-risk questions reach staff
Stop	diagnostic advice, stored health data, or recommendation of stale services

S8 • Equal-seat launch and contribution credit

Carrier/groups	equal-seat forum, adjustable lectern, quiet prep room and contribution-provenance display; women scientists, start-ups, students, carers and frontline maintainers
Space	gaps in prime slots, accessible seats, speaking time and backstage preparation
Algorithm	group gaps in recommendation exposure, speaker order and omitted credit
Data/privacy	voluntarily submitted event materials and human-verified contribution records; anonymous or team credit permitted; no inferred identity attributes
Owner/baseline/window	event curator and contribution-verification group; audit the scheduling template first without claiming a local disparity as fact; 30/60/90 d
Success	every display has verified provenance and seat/time allocation reasons are public
Stop	misattribution, identity marketing, unexplained exposure demotion, or occupied accessible seating

S9 • Education and career recommendation calibration

Carrier/groups	shared learning tables, paper course trees and staffed career advice; girls and women students, returning carers, older learners and account-free visitors
Space	time-slot, quiet-seat, care-compatibility and no-account information gaps
Algorithm	group gaps in field and job exposure, salary bands and advanced-course recommendations
Data/privacy	synthetic learner profiles and public course/job categories; no live education records; no inferred gender or family role; optional self-selected profiles only
Owner/baseline/window	learning-space operator plus education and employment reviewers; human directory choices under a shared task list form the control; 30/60/90 d
Success	mirrored synthetic profiles receive equal field breadth and advancement opportunity
Stop	stereotypes narrow choices, live student records appear, or personalisation cannot be disabled

Twelve dual-scorecard scenarios • 10-12

S10 • Automated shuttle parity experience

Carrier/groups	off-station mock waiting island, staffed guide desk and accessible boarding prototype; night workers, wheelchair or stroller users, slow older walkers and visitors
Space	gaps in waiting, boarding, ramp availability and sheltered seating
Algorithm	group gaps in dispatch wait, detour, denied boarding and takeover time
Data/privacy	simulated services and volunteer tasks; no live public-transport dispatch; no facial ticketing or inferred mobility; anonymised task records
Owner/baseline/window	transport-test operator plus accessibility and safety reviewers; matching-time controls using staffed shuttle and walking alternatives; 30/60/90 d
Success	zero denied boardings and waiting/boarding gaps within preregistered tolerance
Stop	denied boarding, ramp failure, unavailable takeover, or wrong last-service information

S11 • Commercial recommendation and price consistency audit

Carrier/groups	public price interfaces and no-login lookup terminals in the everyday parity arcade; local workers, visitors, older people, carers and multilingual users
Space	gaps in visibility, queue, seating and accessible reach for equivalent offers
Algorithm	mirrored-profile gaps in ranking, discount, price display and event recommendation
Data/privacy	public catalogues, synthetic sessions and manual cross-sectional records; no fingerprinting or cross-shop tracking; clear terminals after every session
Owner/baseline/window	commercial operator and consumer-rights reviewer; same-time, same-offer comparison across synthetic sessions; 30/60/90 d
Success	same-condition prices match and differentiated recommendation is explained and switchable
Stop	hidden differential price, unswitchable tracking, unknown price source or sensitive profiling

S12 • Dual-bias appeal and public retest

Carrier/groups	three calibration desks, phone and paper entry, and quarterly public retest sessions; all users, prioritising affected people and those with low digital skills
Space	distance to appeal, wait, accessibility, resolution time and visibility of spatial repairs
Algorithm	group gaps in intake, human review, rollback and correction outcome
Data/privacy	minimal incident records, public issue categories and anonymised outcomes; separate appeal and operations logs; access, correction and consent withdrawal available
Owner/baseline/window	independent appeal officer, scenario owner and public observers; publish intake criteria, owner, deadline and escalation route before the pilot; 30/60/90 d
Success	100% of appeals have receipt, owner and deadline, with traceable retest results
Stop	appeal affects eligibility, unexplained overdue cases, impossible rollback or retaliation

Concept projects and phasing

ID	Project	Locus	场景	Prerequisite
JZ-F01	Zhongzhiyuan closed dual-bias test yard	PROV-KEY-001	T1-T3	需运营主体、测试伦理、安全审批和企业自愿接入；不进入真实街道
JZ-F02	AI Origin care-time-chain segment	PROV-KEY-002	S6-S9	需现场测绘、设施权属、服务机构和分时基线
JZ-F03	Dazhongsi night-fairness segment	PROV-KEY-003	T4,S10-S11	需站点/道路条件、夜间照度、运营许可和安全复核
JZ-F04	Three dual-bias appeal desks	三处重点区	S12	需独立责任人、期限、数据隔离和版本回滚协议
JZ-F05	Equal-section public-space component library	SITE-001	全场景	需无障碍、消防、文保、绿地及道路专业深化
JZ-F06	Quarterly baseline and annual public retest	运营网络	全场景	需预算、采购条款、公众观察员和匿名发布规范
JZ-F07	Three contribution and calibration honour nodes	三处重点区	S8,S12	仅为概念性展示系统，需文化策展、版权和场地许可
JZ-F08	Fair procurement and scenario admission clauses	治理层	T1-T4	需法务、行业监管与专业团队确认，非已定政策

0-3 months fill data and accountability gaps; 4-12 months run only reversible pilots; 1-3 years deepen only dual-passing and legally approved work; long term publishes open protocols.

Risk, rights and non-claims

01

Misuse of provisional geometry; recalculate fully after official polygons.

02

No identity inference, biometrics, worker scoring or automated live eligibility.

03

Either the safety lead or independent public-interest/ethics lead can stop a trial.

04

External cases are summarised only; visuals are original; Noto Sans SC uses SIL OFL 1.1.

05

No visit, interviews or test; no claim of community support or observed fairness.

06

Submission, merge, score or display is not government adoption, planning approval or construction authority.

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